

DS 3021 HW: Regularization

$$1. J(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n (\tilde{y}_i - (b_0 + b_1 \tilde{x}_i))^2 + a(b_1)^2$$

$$2. \frac{\partial J}{\partial b_1} = -\frac{2}{n} \sum_{i=1}^n \tilde{x}_i (\tilde{y}_i - b_1 \tilde{x}_i) + 2ab_1 = 0$$

$$-\frac{1}{n} \sum \tilde{x}_i \tilde{y}_i + \frac{1}{n} \sum b_1 \tilde{x}_i^2 + ab_1 = 0$$

$$\frac{1}{n} \sum b_1 \tilde{x}_i^2 + ab_1 = \frac{1}{n} \sum \tilde{x}_i \tilde{y}_i$$

$$b_1 \left(\frac{\sum \tilde{x}_i^2}{n} + a \right) = \frac{\sum \tilde{x}_i \tilde{y}_i}{n} \quad n \rightarrow b_1 (\sum \tilde{x}_i^2 + na) = \sum \tilde{x}_i \tilde{y}_i$$

$$b_1 = \frac{\sum \tilde{x}_i \tilde{y}_i}{\sum \tilde{x}_i^2 + na}$$

$$\frac{\partial J}{\partial b_0} = \left(-\frac{2}{n} \sum_{i=1}^n (\tilde{y}_i - b_0 - b_1 \tilde{x}_i) = 0 \right) \cdot \frac{1}{-2}$$

$$= \frac{1}{n} (\sum \tilde{y}_i - \sum b_0 - \sum b_1 \tilde{x}_i) = 0$$

* original problem states that

$$b_0 = \frac{1}{n} \sum \tilde{y}_i - \frac{1}{n} \sum b_1 \tilde{x}_i \rightarrow \frac{1}{n} \sum \tilde{y} = 0 \rightarrow b_0 = 0 - b_1(0)$$

$$\frac{1}{n} \sum \tilde{x} = 0$$

$$b_0 = 0$$